Design and Analysis of Algorithms 18/03/2026

Lecture 16: All Pairs Shortest Path Problem

Instructor: Sourav Chakraborty *Scribe: Muhammad KM, Ankit Yadav*

Problems

Let $G = (V, E, W)$, where $W : E \to \mathbb{R}$ and G contains no negative cycles.

- Given $u, v \in V$, output the shortest path from u to v.
- **Single Source Shortest Path (SSSP):**

  Given $u \in V$, for all $v \in V$, output the shortest path from u to v.
- **All Pairs Shortest Path (APSP):**

  For all $u, v \in V$, output the shortest path from u to v.

Dijkstra's Algorithm

- Single pair shortest path: $O(|V|^2)$
- Single source shortest path: $O(|V|^2)$
- All pairs shortest path: $O(|V|^3)$

Bellman-Ford Algorithm

- Single source shortest path: $O(|V||E|)$
- All pairs shortest path: $O(|V|^2|E|)$

Adjacency Matrix and Path Lengths

Let A be the adjacency matrix of a directed unweighted graph.

- $(A^{\ell-1})_{u,v} = 0$ if there is no path of length $\ell - 1$ from u to v

- $(A^\ell)_{u,v} > 0$ implies existence of a path of length ℓ

Dynamic Programming Approach

- Break the problem into smaller subproblems
- Store solutions in a table
- Combine solutions to build larger solutions

Subproblem for APSP (Repeated Squaring)

$D^\ell(u,v) =$ length of shortest path from u to v using at most 2^ℓ edges

$$D^0(u,v) = \begin{cases} W(u,v) & (u,v) \in E \\ \\ +\infty & \text{otherwise} \end{cases}$$

$$D^\ell(u,v) = \min_{w \in V} \left\{ D^{\ell-1}(u,w) + D^{\ell-1}(w,v) \right\}$$

Algorithm 1 All-Pairs Shortest Paths (Repeated Squaring)

for all $u, v \in V$ **do**
    $D^0(u,v) = \begin{cases} W(u,v) & (u,v) \in E \\ +\infty & \text{otherwise} \end{cases}$
end for
for $\ell = 1$ to $\lceil \log |V| \rceil$ **do**
    for all $u, v \in V$ **do**
        $D^\ell(u,v) = \min_{w \in V} \left\{ D^{\ell-1}(u,w) + D^{\ell-1}(w,v) \right\}$
    end for
end for

Floyd-Warshall Algorithm

Let vertices be numbered as $v_1, v_2, \ldots, v_n$.

$D^\ell(u,v) =$ shortest path from u to v using only $\{v_1, \ldots, v_\ell\}$ as intermediate vertices

$$D^\ell(u,v) = \min \left\{ D^{\ell-1}(u,v),\ D^{\ell-1}(u,v_\ell) + D^{\ell-1}(v_\ell,v) \right\}$$

- Either the shortest path from i to j does not use vertex k
- Or it passes through k (i.e., $i \to k \to j$)

Required Conditions

- The graph must be a weighted graph (directed or undirected)
- Edge weights may be positive or negative
- The graph must not contain any negative weight cycles

Algorithm 2 Floyd-Warshall Algorithm

for all $u, v \in V$ **do**
$$D^0(u,v) = \begin{cases} W(u,v) & (u,v) \in E \\ +\infty & \text{otherwise} \end{cases}$$
end for
for $\ell = 1$ to $|V|$ **do**
  for all $u, v \in V$ **do**
$$D^\ell(u,v) = \min\left\{D^{\ell-1}(u,v),\ D^{\ell-1}(u,v_\ell) + D^{\ell-1}(v_\ell,v)\right\}$$
  end for
end for

Example: Floyd-Warshall Algorithm

Consider the graph:

$$V = \{1, 2, 3\}$$

Weight matrix:

$$D^0 = \begin{bmatrix} 0 & 3 & 10 \\ \infty & 0 & 1 \\ \infty & \infty & 0 \end{bmatrix}$$

Interpretation:

- Direct edge $1 \to 2$ has weight 3
- Direct edge $1 \to 3$ has weight 10
- Direct edge $2 \to 3$ has weight 1

- No other direct edges exist

Goal: Find shortest paths between all pairs using intermediate vertices.

Iteration $\ell = 1$: Allow vertex 1 as intermediate

We check if any path improves by going through vertex 1.

For every pair (u, v):

$$D^1(u, v) = \min\left(D^0(u, v),\ D^0(u, 1) + D^0(1, v)\right)$$

Key observations:

- No vertex has a path to 1 (except 1 itself)
- So no new shorter paths can be formed via 1

Result:

$$D^1 = D^0$$

Iteration $\ell = 2$: Allow vertex 2 as intermediate

Now we allow paths that go through vertex 2.

$$D^2(u, v) = \min\left(D^1(u, v),\ D^1(u, 2) + D^1(2, v)\right)$$

Important computation:

For $(u, v) = (1, 3)$:

$$\text{Direct path: } D^1(1, 3) = 10$$

$$\text{Via 2: } D^1(1, 2) + D^1(2, 3) = 3 + 1 = 4$$

$$D^2(1, 3) = \min(10, 4) = 4$$

All other pairs do not improve.

Updated matrix:

$$D^2 = \begin{bmatrix} 0 & 3 & 4 \\ \infty & 0 & 1 \\ \infty & \infty & 0 \end{bmatrix}$$

Iteration $\ell = 3$: Allow vertex 3 as intermediate

$$D^3(u,v) = \min\left(D^2(u,v),\ D^2(u,3) + D^2(3,v)\right)$$

Observation:

- Vertex 3 has no outgoing edges to other vertices
- So no shorter paths can be formed through 3

Result:

$$D^3 = D^2$$

Final Shortest Path Matrix:

$$\begin{bmatrix} 0 & 3 & 4 \\ \infty & 0 & 1 \\ \infty & \infty & 0 \end{bmatrix}$$

Final Interpretation:

- Shortest path $1 \to 3$ is now $1 \to 2 \to 3$ with cost 4
- All other shortest paths remain unchanged

Summary of APSP Complexities

- Dijkstra (non-negative weights): $O(|V|^3)$
- Bellman-Ford (with negative weights): $O(|V|^2|E|)$
- Repeated Squaring: $O(|V|^3 \log |V|)$
- Floyd-Warshall: $O(|V|^3)$

Uses Cases:

- If all weights are non-negative: use Dijkstra
- If negative weights exist: use Bellman-Ford or Floyd-Warshall
- If all-pairs shortest paths required: use Floyd-Warshall